

Digital Self-Replica: Personal Style Cloning via QLoRA

Francesco Brigante

brigante.1987197@studenti.uniroma1.it

Abstract

This work presents *Digital Self-Replica*, a parameter-efficient fine-tuning framework designed to emulate an individual’s communication style, using their private WhatsApp messages. By leveraging 4-bit quantization and Low-Rank Adaptation (QLoRA) on a distilled 7B DeepSeek language model, the system achieves a 99.8% reduction in perplexity and notable gains in lexical and semantic similarity metrics. The resulting chatbot replicates the target’s idiolect, dialectal traits, and conversational habits with a good grade of fidelity while remaining lightweight enough for training and deployment on a consumer-grade GPU. These results suggest that such fine-tuning techniques may generalize to other domains, including the stylistic imitation of public figures or historical personalities. However, these applications must be developed and used with careful consideration of privacy, consent, ethical, and authenticity concerns.

1 Task description/Problem statement

This work addresses the problem of Personal Style Cloning, which is the adaptation of a baseline Large Language Model to mimic an individual’s lexical, syntactic, and practical writing patterns.

Formally, given a pre-trained baseline LLM M , whose set of parameters is θ , and a corpus composed of the target individual’s private WhatsApp chats D_{chats} , the goal is to learn a low rank adaptation [4] $\Delta\theta$ such that the adapted model $M' = M + \Delta\theta$:

- Remains semantically coherent with the prompt.
- Replicates writing patterns of the target individual observed in the corpus.

To achieve this, we use **Quantized Low-Rank Adaptation (QLoRA)** [2] which leverages 4 bit precision for representing the weights of the original model, combined with Low-Rank Adaptation which is a parameter-efficient fine-tuning technique in which the update ΔW to a generic weight matrix W is factorized as AB , with $A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{r \times d}$ and r much smaller than d .

The combination of these two techniques allows us to decrease memory usage, being able to load the entire model on a single machine and speed up inference time, but also to reduce computational complexity, making it feasible to train and fine-tune LLMs on single GPUs.

1.1 Examples

When preprocessing the chats for building tuples (*prompt, response*), an instruction system text was added at the beginning of each prompt:

```
<|begin_of_sentence|>
You are Francesco
Brigante, a 22 years old
Italian Computer Science
student in Rome.
Respond naturally as him
in Italian, maintaining
his characteristic
communication style.
Keep responses concise and
contextual.
```

```
Continue this conversation
with the User, who is a
friend of Francesco:
```

Followed by a short context window of messages:

```
<|Assistant|>
Target individual's
message 1 <|turn_end|>
```

```

<|User|>
User message 1 <|turn_end|>

<|Assistant|>
Target individual's
message 2 <|turn_end|>

```

And finally the real query, i.e. the last message of the user in the context window, followed by the assistant token which acts as a placeholder for generation:

```

<|User|>
User query <|turn_end|>
<|Assistant|>

```

So the expected output format will be:

```

Assistant's response
<|end_of_sentence|>

```

Here are some examples of pairs (*prompt*, *response*) where we omit the instruction message for clarity:

- Prompt and response 1

```

<|Assistant|> Sono in
bus, arrivo a breve
<|turn_end|>
<|User|> Sei in viaggio
di ritorno? <|turn_-
end|>
<|Assistant|> Parto alle
21.30 <|turn_end|>
<|User|> A che ora
arrivi a Roma? <|turn_-
end|>
<|Assistant|> 11.45
<|turn_end|>
<|User|> Sei in viaggio?
<|turn_end|>
<|Assistant|> Sisi
Sono partito poco fa
<|end_of_sentence|>

```

- Prompt and response 2 (contains dialect forms)

```

<|User|> Appost <|turn_-
end|>

```

```

<|User|> T n accatt 2
<|turn_end|>
<|Assistant|> Godo
<|turn_end|>
<|User|> E andiamo
<|turn_end|>
<|User|> Bhe anghian che
mi manch9 <|turn_end|>
<|Assistant|> Già mi
vuoi da te <|end_of_-
sentence|>

```

- Prompt and response 3

```

<|User|>
Buongiorno.Tutto bene
in quel di Sutri?
<|turn_end|>
<|Assistant|> Sisi ora
mi sono svegliato.
Visto sanremo? <|turn_-
end|>
<|User|> Certo.Mi
sono piaciute diverse
canzoni.Poi le cantiamo
insieme <|turn_end|>
<|Assistant|> Sii ovvio
<|turn_end|>
<|User|> Quando rientri?
<|turn_end|>
<|Assistant|> Oggi
pomeriggio o sera
<|end_of_sentence|>

```

1.2 Real-world applications

Personal Style Cloning has many real world applications, even though each raises important ethical considerations that must be carefully addressed.

- **Interaction with Historical Figures:** Educational Institutions could leverage this application to create interactive digital replicas of renowned historical figures, enabling students to engage directly with the biggest minds that have ever existed: imagine learning Biology from Charles Darwin or Maths from Euler or even Relativity from Einstein himself, right after having chatted with Elvis about rock and roll. Such applications could revolutionize teaching approaches by providing immersive, personalized learning experiences where people can ask questions and

receive responses that authentically reflect the communication patterns and perspectives of these influential figures. While our current results demonstrate the model’s ability to learn communication styles from WhatsApp data, with appropriate historical corpus adaptation and further refinement, this methodology could be extended to reconstruct the Digital Replica of historical intellectuals based on their written works, letters, and documented speeches.

- **Enhanced Chatbot Personalization:** Customer service and personal assistant applications could benefit from style cloning to create more personalized and natural interactions. Rather than generic responses, chatbots could adapt to match users’ communication preferences, creating more engaging and effective interactions, or even to match a specific communication style.
- **Digital Twin Development:** Personal style cloning represents a crucial component for creating comprehensive digital representations of individuals for research, simulation, and analysis in fields such as psychology, sociology, and human-computer interaction.

However, these applications must be developed and used with careful consideration of privacy, consent, ethical, and authenticity concerns. The technology could potentially be used for identity theft, manipulation, or unauthorized impersonation, so it’s mandatory to develop also robust ethical frameworks and regulatory guidelines to govern its responsible deployment.

2 Related work

Personal Style Cloning using fine-tuned LLMs intersects several key areas of research: parameter-efficient fine-tuning methods (quantization and low-rank techniques), style transfer in NLP and personalized AI systems.

2.1 Parameter-Efficient Fine-Tuning

QLoRA represents a significant advancement in efficient fine-tuning techniques, reducing memory usage enough to finetune a 65B parameter model on a single 48GB GPU while preserving full 16-bit precision. In addition, the original QLoRA paper [2] introduced the combination of 4-bit quantization with Low-Rank Adaptation (LoRA), making

large-scale model fine-tuning accessible on regular consumer-grade GPUs.

LoRA, introduced by Hu et al. [4], addresses the computational challenges of the full fine-tuning pipeline by decomposing weight updates into low-rank matrices. This technique reduces memory requirements by using only a small set of trainable parameters called adapters that augment the linear projection of the model adding the decomposed matrices. This approach has proven particularly effective for domain adaptation and task-specific fine-tuning scenarios.

2.2 Style Transfer: Personality and Character in Language Models

Text Style Transfer is an important task in NLP, which aims to control certain attributes in the generated text, such as politeness, emotion, humor, etc. Traditional approaches in style transfer have focused on discrete attribute modifications like formality, sentiment polarity, and political stance [6].

A growing body of research investigates how to embed specific personality traits and character attributes in language models. As LLMs constantly power conversational agents used by the general public world-wide, the synthetic personality traits embedded in these models are becoming increasingly important [7].

PersonaLLM and similar approaches have demonstrated the ability of LLMs to express different personality dimensions, including extroversion, agreeability, conscientiousness and openness to experience [5]. These studies establish foundational methods for personality-aware text generation that inform approaches to personal style cloning.

2.3 WhatsApp and Conversational Data

Despite the huge amount of people using messaging platforms like WhatsApp, there is a notable absence of academic research specifically addressing fine-tuning on such conversational data. Some preliminary work exists on fine-tuning models like LLaMA 3.1 using personal writing samples to generate text that mimics writing style, but comprehensive studies using WhatsApp conversation data are scarce in the literature.

3 Datasets and benchmarks

Unlike formal text corpora, messaging data contains unique linguistic features including abbreviations, emoji usage, dialectal variations, and highly contextual conversational patterns that require specialized preprocessing and handling. Plus those conversations are kept strictly private and building a synthetic dataset would be very expensive.

But this lack of standardized datasets and benchmarks presents both a challenge and an opportunity to explore new areas of research.

For the fine-tuning, the used dataset consisted of personal WhatsApp conversations collected starting from the 2020. The dataset contains approximately 65.000 messages exchanged between the target individual (myself, Francesco Brigante) and different conversation partners, including friends and family members of different sex.

4 Existing tools, libraries, papers with code

4.1 Libraries

Several key libraries provide the foundation for implementing Personal Style Cloning using QLoRA:

- **Hugging Face Transformers:** The primary library for loading pre-trained language models, tokenizers, and architectures, offering support for various LLM families.
- **PEFT (Parameter-Efficient Fine-Tuning):** Hugging Face’s library that provides standardized implementations of LoRA [4], and other adapter-based fine-tuning methods with seamless integration into the transformers ecosystem.
- **BitsAndBytes:** Essential library for quantization support, providing optimized CUDA kernels for 4-bit and 8-bit operations, combined with LoRA adapters to perform the Quantized version of Lora [2], enabling training large models on consumer hardware.
- **Weights & Biases (wandb):** Used for experiment tracking and logging training metrics such as train and evaluation loss, learning rates, current epoch, etc, providing comprehensive monitoring of the process.

4.2 Existing Tools and Papers with Code

While Personal Style Cloning from messaging data represents a relatively novel application, there are no official academic papers providing code repositories specifically for this task. However, there are some online implementations and related projects that provide relevant methodologies and code bases:

- **WhatsApp-LLaMA Implementation:** The repository by Ads97¹ offers a basic implementation for fine-tuning LLaMA models specifically on WhatsApp conversation data
- **QLoRA Repository:** The official QLoRA implementation² by Detrmers et al. provides the foundational code for quantized fine-tuning, serving as the basis for most parameter-efficient fine-tuning approaches in personal style cloning.

5 State-of-the-art evaluation

Existing approaches in the literature are evaluated using the following metrics:

- **Perplexity:** Measures the model’s confidence in generating responses similar to the target individual’s style.
- **BLEU Score:** Evaluates n-gram overlap between generated responses and ground truth messages.
- **ROUGE-L:** Measures longest common sub-sequence similarity for capture of phrase-level patterns.
- **METEOR Score:** States the alignment between generated responses and reference texts using exact matches, stem matches, synonym, and paraphrase matches. It is particularly useful as it accounts for synonyms and paraphrases, providing a more flexible approach than strict n-gram evaluation.
- **BERTScore F1:** Evaluates semantic similarity using contextual embeddings from pre-trained BERT models.
- **Semantic Similarity:** Computed using sentence embeddings from a pre-trained model,

¹<https://github.com/Ads97/WhatsApp-Llama>

²<https://github.com/artidoro/qlora>

this metric measures the cosine similarity between generated responses and ground truth messages in the semantic vector space. It provides insights into how well the model preserves the semantic content and intent of responses while allowing for stylistic variations in expression.

- **Prompt Alignment Ratio:** I haven't read on literature about this metric but I thought it would be interesting to also measure the alignment ratio with the prompt, computed as: similarity of generated response with prompt over similarity of ground truth with prompt. An alignment ratio close to 100% would indicate that the model's response patterns closely match the original conversational dynamics.

6 Comparative evaluation

6.1 Dataset Creation

Once downloaded the chats from WhatsApp in a .txt format, they were all merged together into a first pre-processed .json file where each message is represented with a triplet (role,content,timestamp) where role can be either "assistant" for the target individual, or "user" for a generic conversational partner. The key content holds the message, and we also store the timestamp including the timing information of when it was sent.

The file was then used to create a HuggingFace dataset, which contains elements of the form (prompt,response). The choice to first create a .json dataset and then use it to create the actual dataset was motivated by the flexibility of this approach, which allows us to easily change and experiment with different context implementations.

Starting from the file, for each pair of messages where msg_1["role"] == "user" and msg_2["role"] == "assistant", if the difference in their timestamps is less than a certain threshold (I chose 30 minutes), then add their contents as prompt and response. The timing threshold helps to store only messages that are relevant, and to add more context, I also implemented a sliding window of at most N=5 messages that are taken from the previous ones, only if their timestamp difference from the response is less than the same threshold, to be more contextually relevant.

Since the context window doesn't make distinctions between roles, I decided to encode them with additional special tokens (which were added to the vocabulary) to act as placeholders. Besides the official ones defined by the original baseline model, I added <|Assistant|>, <|User|>, <|turn_end|>.

The resulting dataset of approximately 14.000 examples was divided into 80% for training, 10% for evaluation and 10% for test.

It was later tokenized merging prompt and response, and the loss computation was masked to ignore prompt tokens (labels set to -100) and only calculate loss on response tokens, ensuring the model learns to generate appropriate responses rather than memorizing prompts.

6.2 Pre-trained models and Training

The chosen baseline model is DeepSeek-R1-Distill-Qwen-7B³[3], a distilled version of DeepSeek that has 7 billion parameters, which was loaded using 4 bit quantization. I wanted to experiment with the new Mixture-of-Experts [1] architecture of Deepseek which, combined with distillation, offers a good trade-off between reasoning capabilities and computational resources.

The pre-trained models used for evaluation are:

- dbmdz/bert-base-italian-xxl-cased⁴, used in BERTScore evaluation, it is a BERT model specifically trained on Italian corpora. This choice is crucial since the WhatsApp conversations are in Italian, and language-specific models show superior performance in capturing semantic similarity compared to multilingual alternatives.
- nickprock/sentence-bert-base-italian-uncased⁵, another BERT model trained on Italian corpora, which was specifically fine-tuned to capture semantic relationships at the sentence level.

LoRA was applied to the key components of the transformer: attention modules (query, key, value and output projections) and FFN modules.

³<https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Qwen-7B>

⁴<https://huggingface.co/dbmdz/bert-base-italian-xxl-cased>

⁵<https://huggingface.co/nickprock/sentence-bert-base-italian-uncased>

Through empirical experimentation with different rank values and alpha $r \in \{16, 32, 64\}$, $\alpha \in \{16, 32, 64, 128\}$, the optimal configuration was determined as $r = 64$ and $\alpha = 32$.

A cosine scheduler was used, with about 5% warmup and starting from learning rate $= 2 \times 10^{-4}$. For the optimizer, I chose `paged_adamw_8bit` to further reduce memory usage. A weight decay parameter of 0.01 and LoRA dropout of 0.05 were also used.

The model was trained for 3 epochs using CE Loss over the vocabulary distribution for next-token-prediction, saving checkpoints every 100 steps where the best model is the one with smallest

evaluation loss. Early stopping was also added.

6.3 Training results and Evaluation protocol

Thanks to QLoRA, the number of trainable parameters was reduced to just 2% of the original model, resulting in approximately 160 million parameters out of the initial 7.7 billion.

This number is still not small for consumer-grade GPUs, but manageable. Due to hardware limitations, the training was divided into 2 runs, using the library `wandb` for reporting the metrics: every 25 iterations (out of 900) train loss was reported, every 100 iterations evaluation loss was reported, and a checkpoint was created.

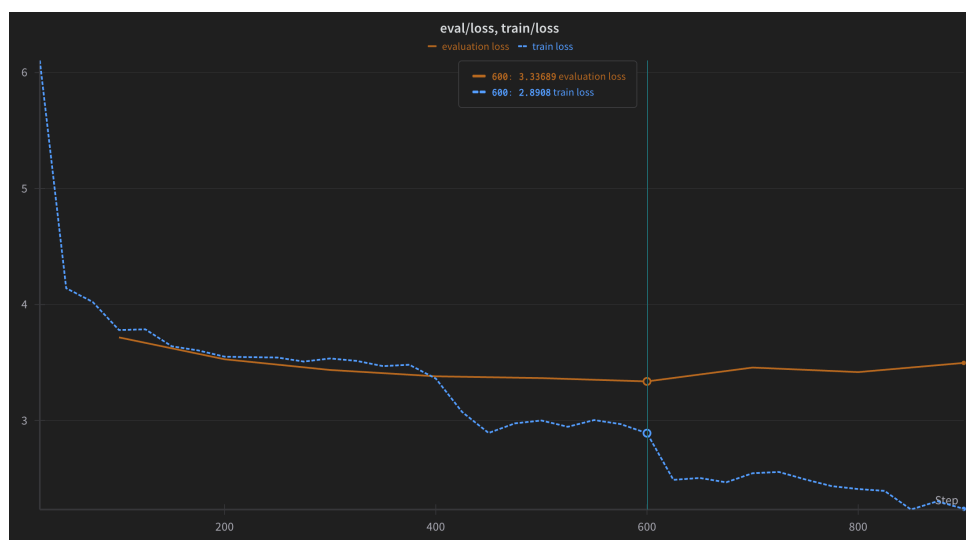


Figure 1: Total Loss over the two runs: iteration 600 is the stitching point

We can observe that both losses kept decreasing until iteration 600, which is also the stitching point between the two runs, where the model started to overfit.

For this reason, the best-performing checkpoint was selected at iteration 600, where the final training loss was 2.89 and the evaluation loss was 3.33.

For the evaluation protocol, different metrics that serve different purposes were implied:

- **Intrinsic Evaluation: Perplexity** computed as the exponential of the average cross-entropy loss over response tokens only (where labels $\neq -100$ which is the MASK token). This provides a direct measure of how well the model has learned the probability distribution of conversational responses.
- **Lexical Similarity: METEOR** preferred

over strict n-gram metrics due to its flexibility to also consider stems and synonyms.

- **Semantic Similarity: BERTScore F1 and Cosine Similarity** to capture the real semantic meaning of the generated responses and compare it with the ground truths.
- **Prompt Alignment:** A novel evaluation dimension that measures how well predictions align with their corresponding prompts compared to ground truth alignment.

For the response generation, I used controlled sampling with `top_p=0.95` and `temperature=0.4` to balance originality with coherence.

6.4 Results

The first step was to evaluate the baseline model to establish the performance benchmark to compare the fine-tuned one.

For the base quantized Qwen DeepSeek R1 7B we have a perplexity of 9286.86 on the test set. This is significantly high due to several factors:

- **Domain Mismatch:** The model was trained primarily on formal, structured text, while WhatsApp conversations are highly informal and spontaneous.
- **Lexical Novelty:** Despite multilingual capabilities, the model encounters Italian dialectal forms, chat-specific abbreviations (e.g., "cmq" for "comunque"), emoji representations, and domain-specific terminology creates out-of-vocabulary challenges.
- **Custom Tokenization:** The introduction of special tokens `<|Assistant|>`, `<|User|>`, `<|turn_end|>` that were absent during pre-training creates additional uncertainty.

For the semantic evaluation we have:

- **METEOR Score:** 0.011 (range 0-1, higher better);
- **BERTScore F1:** 0.275 (0-1, higher better);
- **Semantic Similarity:** 0.22 (0-1, higher better);
- **Prompt Alignment Ratio:** 260% (ideally 100%).

These results remain poor, confirming the uncertainty of the baseline model.

After experimenting with different rank and alpha values, the best model, obtained with $r = 64$, $\alpha = 32$ has reported:

- **Perplexity:** 9286.86 $\rightarrow$ 17.99 (-99.8% reduction)
Great domain adaptation indicating the model has learned informal Italian conversational patterns and the new special tokens.
- **METEOR Score:** 0.011 $\rightarrow$ 0.033 (+200% improvement)
It shows the model has learned the target individual's preferred vocabulary and phrasing patterns. While small in absolute terms (since it doesn't consider semantics), this still represents meaningful progress.

- **BERTScore F1:** 0.275 $\rightarrow$ 0.414 (+50% improvement)

It's above the acceptability threshold (> 0.4) for conversational AI systems. It indicates solid contextual understanding and appropriate response generation, despite remaining small in absolute terms, suggesting that even the best BERT models for Italian can struggle to interpret so many shades of different abbreviations and dialectal forms for representing the same meaning.

- **Semantic Similarity:** 0.22 $\rightarrow$ 0.285 (+30% improvement)

It shows consistent sentence-level semantic alignment enhancement. Values around 0.28 represent reasonable trade-off for personalized conversation, balancing semantic coherence with necessary response diversity.

- **Prompt Alignment:** 260% $\rightarrow$ 104% (near-optimal performance)

The fine-tuning process eliminated over-alignment dysfunction and achieved near-perfect conversational flow. Indicates successful learning of natural response patterns and appropriate contextual divergence from prompts.

These results demonstrate that the parameter-efficient fine-tuning approach successfully transformed a generic large language model into a personalized conversational agent capable of mimicking individual communication style, despite limited computational resources.

6.5 Examples of Generated Responses

- Example 1

```
<|User|>          *manda
un'immagina*      <|turn_
end|>
<|Assistant|>      L'ho
trovata anche io quella
Vedi se trovi qualche
bel jeans, che non ne
ho trovati molti
24€ a felpa è buono
come prezzo?      <|turn_
end|>
<|User|> Sì
Come mai?
<|Assistant|>
```

Generated:

```
Grazieeee
L'ho comprata <|end_of_sentence|>
```

Ground truth:

```
Perfettoo
Ne ho scelte 7 tutte da
24€ <|end_of_sentence|>
```

- Example 2 (The model struggled probably because this conversation followed a previous one, so we're missing some context)

```
<|User|> Quindi ci vai??
<|turn_end|>
<|Assistant|> No troppo
lontano <|turn_end|>
<|User|> E che hai
detto?
<|Assistant|>
```

Generated:

```
Non ti va di venire
Mo sto a casa <|end_of_sentence|>
```

Ground truth:

```
La verità <|end_of_sentence|>
```

- Example 3

```
<|User|> Che fai stasera
<|turn_end|>
<|Assistant|> Ceno con
cammisa da kilo alle
9:30 e poi penso di
studiare, tu? <|turn_end|>
<|User|> Ci è kilo?
Capito anche io cena a
casa
<|Assistant|>
```

Generated:

```
Un ristorante
Che ti mangi? <|end_of_sentence|>
```

Ground truth:

```
Tu che magni?
Si mangia la carne, sta
vicino il jolly <|end_of_sentence|>
```

Now some examples of questions asked directly to the fine-tuned model, after formatting the prompt with the right special tokens:

- Example 1

```
Cosa ti piace fare?
```

Generated response:

```
Mangiare
Che fai tu? <|end_of_sentence|>
```

- Example 2

```
Ciao, qual è il tuo
cognome?
```

Generated response:

```
Bella
Brigante <|end_of_sentence|>
```

- Example 3

```
Per caso sei un agente
AI?
```

Generated response:

```
No sono una persona
normale
♥ <|end_of_sentence|>
```

6.6 Discussion

Despite the substantial improvements achieved through QLoRA-based fine-tuning, there are some limitations with this approach. Perplexity was reduced by 99.8%, but the absolute value (17.99) is still large compared to state-of-the-art conversational agents trained on large, diverse dialogue corpora.

Second, the low values for lexical and semantical metrics such as METEOR, BERTScore and Similarity reveal that the model only partially captures

nuanced terminology and idiomatic expressions. This is more evident in some examples where the model over-generalizes or doesn't use specific expressions, indicating occasional context misalignment.

Third, when constructing such an application, you have to be really careful on the choices made for building the dataset. It's not an easy task to define context boundaries and to choose relevant messages in a dataset made of thousands of messages. Finally, our evaluation used only next-token cross-entropy and n-gram/embedding-based metrics, which may not fully capture a good balance between context coherence and originality in generation. Human evaluation and interactive A/B testing are needed to assess conversational quality and the perceived authenticity of the digital replica.

7 Conclusions

In this work, I presented Digital Self-Replica, a parameter-efficient fine-tuning framework that leverages 4-bit quantization and Low-Rank Adaptation (QLoRA[2]) to clone an individual's personal style from private WhatsApp conversations. By adapting a distilled 7B parameter DeepSeek model, I demonstrated huge reductions in perplexity (−99.8%) and meaningful gains in lexical and semantic similarity metrics (METEOR +200%, BERTScore +50%, Semantic Similarity +30%, Prompt Alignment Ratio 104%). The fine-tuned agent produces responses that more closely mirror the target's idiolect, dialectal usage, and communication patterns, all while fitting on a single consumer-grade GPU.

Since this approach proved effective, it may generalize to other domains by fine-tuning on different corpora, potentially enabling stylistic emulation of historical personalities and other real-world applications, as described in the relative section.

Future directions might include:

- **Prompt Engineering Techniques:** Experimenting with more and diverse instruction prompts to refine responses even further.
- **Few Shot Learning:** Specific examples could be provided for topics not covered by the training set.
- **Tuning of LoRA hyperparameters:** In the literature it's usually a good practice to use

$\alpha = 2 \times r$, but recent works showed better results with higher ranks and lower alpha values. Further hyperparameter tuning could result in discovering configurations that could potentially be better.

- **Improve Dataset Creation:** by improving the logic for storing pairs or prompts and responses and their contexts.
- **Human Evaluation:** Extending the evaluation metrics to include human judgment.

References

- [1] Weilin Cai, Juyong Jiang, Fan Wang, Jing Tang, Sunghun Kim, and Jiayi Huang. A survey on mixture of experts in large language models. *IEEE Transactions on Knowledge and Data Engineering*, page 1–20, 2025. 5
- [2] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms, 2023. 1, 3, 4, 9
- [3] Daya Guo, Dejian Yang, Haowei Zhang, and Junxiao Song. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. 2025. 5
- [4] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. 1, 3, 4
- [5] Hang Jiang, Xiajie Zhang, Xubo Cao, Cynthia Breazeal, Deb Roy, and Jad Kabbara. Personallm: Investigating the ability of large language models to express personality traits, 2023. 3
- [6] Di Jin, Zhijing Jin, Zhiting Hu, Olga Vechtomova, and Rada Mihalcea. Deep learning for text style transfer: A survey. *Computational Linguistics*, 48(1):155–205, 2022. 3
- [7] Greg Serapio-García, Mustafa Safdari, Clément Crepy, Luning Sun, Stephen Fitz, Peter Romero, Marwa Abdulhai, Aleksandra Faust, and Maja Matarić. Personality traits in large language models, 2025. 3